

REPLAY RUN RECORD

Urban Corridor — replay run 2f7a91

Replay Plan rp-114, compiled from scenario **urban-corridor v7**. Hardware binding was resolved at compile time; the allocation below is the record of which physical TX channel carried which drone signal, and how coherent groups were formed for the direction-finding receivers.

RUN STATE RUNNING 02:14 / 10:00	TX CHANNELS 9 active 1 reserved · 2 coherent groups	SHORTEST LEASE 01:48 CG-1 · AG-01 ch2/ch3	SAFETY EVENTS 1 critical 2 warnings this run
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Allocation model

- ☐ **Detection-only receivers — one drone per channel.** One recorded signal is replayed on one physical TX channel; channels are independent and leased individually.
- ☐ **AOA / TDOA receivers — parallel coherent channels.** One drone is prepared across several channels at once — one per array element or TDOA node, carrying authored $\Delta\phi$ / τ offsets — and the channels are combined on the RF network. The group is allocated, leased, armed and muted as a single unit; a partial group is not a valid allocation.
- ☒ **Planned extension.** Combine multiple recordings at digital level and stream several drones per channel — a channel then carries N signal mappings instead of one, while coherent groups stay group-allocated.

CHANNEL	GROUP	SIGNAL MAPPING	FC / SPAN	OFFSET	STATE	LEASE / POWER
CH A0	DETECT	L1 · C2 — Perimeter Patrol Alpha	2.4125 GHz · 20 MHz	independent	TX	03:12 · -12 dBFS
CH A1	DETECT	L2 · VIDEO — Perimeter Patrol Alpha	5.8000 GHz · 61 MHz	independent	TX	03:12 · -6 dBFS
CH A2	CG-1 · TDOA 1/2	L3 · TELEMETRY — Grid Search Bravo → RX-2a	915.0 MHz · 4 MHz	τ 0 ns (ref)	TX	01:48 · -18 dBFS
CH A3	CG-1 · TDOA 2/2	L3 · TELEMETRY — Grid Search Bravo → RX-2b	915.0 MHz · 4 MHz	τ +214 ns	TX	01:48 · -18 dBFS
CH B0	DETECT	L4 · DATA — Grid Search Bravo (ext trigger)	2.4520 GHz · 15 MHz	independent	ARMED	01:48 · —
CH S0	CG-2 · AOA 1/4	L5 · VIDEO — Orbit Charlie → RX-3 el 1	5.7800 GHz · 40 MHz	$\Delta\phi$ 0°	TX (sim)	∞ sim · -3 dBFS
CH S1	CG-2 · AOA 2/4	L5 · VIDEO — Orbit Charlie → RX-3 el 2	5.7800 GHz · 40 MHz	$\Delta\phi$ 63°	TX (sim)	∞ sim · -3 dBFS
CH S2	CG-2 · AOA 3/4	L5 · VIDEO — Orbit Charlie → RX-3 el 3	5.7800 GHz · 40 MHz	$\Delta\phi$ 126°	TX (sim)	∞ sim · -3 dBFS
CH S3	CG-2 · AOA 4/4	L5 · VIDEO — Orbit Charlie → RX-3 el 4	5.7800 GHz · 40 MHz	$\Delta\phi$ 189°	TX (sim)	∞ sim · -3 dBFS
CH B1	SPARE	unmapped — spare allocation	2.4370 GHz · 20 MHz	—	RESERVED	01:48 · —

Receivers and the channels they require

RECEIVER	ROLE	SITE	CHANNELS REQUIRED
RX-1	MONITOR	fixed mast · 12 m AGL · omni	detection only — 1 channel per drone signal (CH A0, A1, B0)
RX-2	TDOA	TDOA node · PPS locked · 8 m AGL	2 parallel channels, time-aligned — CG-1 (CH A2 / A3), τ offsets per node
RX-3	AOA_DOA	array group · 4 elements · Δ 0.06 m	4 phase-coherent channels — CG-2 (CH S0–S3), $\Delta\phi$ per element

Hardware inventory — physical TX channels

CH	AGENT	DEVICE	FREQ RANGE	SAMPLE RATE	LEASE HOLDER	HEALTH
A0	AG-01	X440 · sn 3141A	30 MHz – 4 GHz	0.2 – 500 MS/s	rp-114	nominal
A1	AG-01	X440 · sn 3141A	30 MHz – 4 GHz	0.2 – 500 MS/s	rp-114	nominal
A2	AG-01	X440 · sn 3141A	30 MHz – 4 GHz	0.2 – 500 MS/s	rp-114 · CG-1	nominal
A3	AG-01	X440 · sn 3141A	30 MHz – 4 GHz	0.2 – 500 MS/s	rp-114 · CG-1	nominal
B0	AG-02	AIR7311 · sn 77C2	0.3 – 6 GHz	1 – 125 MS/s	rp-114	nominal
B1	AG-02	AIR7311 · sn 77C2	0.3 – 6 GHz	1 – 125 MS/s	rp-114 (reserved)	thermal 71 °C
C0	AG-03	X440 · sn 3199B	30 MHz – 4 GHz	0.2 – 500 MS/s	—	clock fault
S0	AG-04	sim-dev-0	0.05 – 8 GHz	0.1 – 200 MS/s	rp-114 · CG-2	nominal
S1	AG-04	sim-dev-0	0.05 – 8 GHz	0.1 – 200 MS/s	rp-114 · CG-2	nominal
S2	AG-04	sim-dev-1	0.05 – 8 GHz	0.1 – 200 MS/s	rp-114 · CG-2	nominal
S3	AG-04	sim-dev-1	0.05 – 8 GHz	0.1 – 200 MS/s	rp-114 · CG-2	nominal

Agents

AGENT	HOST / FAMILY	STATUS	CLOCK	LEASE
AG-01	rogue-sdr-a01.lab · Ettus X440	online	GPSDO locked · 1e-11	rp-114 (4/4 ch · incl CG-1)
AG-02	rogue-sdr-a02.lab · Deepwave AIR7311	online	PTP holdover 42 s	rp-114 (2/2 ch)
AG-03	rogue-sdr-b01.lab · Ettus X440	degraded	ref unlocked	—
AG-04	sim-agent-01.svc · Simulated	online	virtual clock	rp-114 (4/8 ch · CG-2)
AG-05	rogue-sdr-c01.lab · Deepwave AIR7311	offline	—	—

Watchdog & safety feed

TIME	SEVERITY	SOURCE	MESSAGE
02:12.4	CRITICAL	watchdog	No-transmit violation predicted: M3 trajectory enters North Ridge in 18 s — L5 gate will mute CH S0.
02:04.1	WARNING	AG-02	Clock degradation: PTP holdover 42 s. TDOA accuracy budget exceeded on RX-2 pairs.
01:58.8	WARNING	AG-02/ch1	Buffer underrun ×3 on CH B0 during ARM idle. Sample pump requested.
01:41.0	INFO	plan rp-114	Channel switch executed: L1 2412 → 2437 MHz on CH A0 (scripted @ t+120).
01:22.7	INFO	lease	Lease extended: AG-01 ch0–ch3 +240 s (holder rp-114) — coherent group CG-1 leased as one unit.
00:47.2	WARNING	AG-03	Agent degraded — reference unlocked. Excluded from allocation; run continues with 10 channels.
00:11.5	INFO	run	Run state ARMED → RUNNING. 10 physical TX channels allocated (4 detection-only, 2 in TDOA group CG-1, 4 in AOA group CG-2), 1 reserved.
00:00.0	INFO	run	Replay Plan rp-114 compiled from urban-corridor v7. Hardware binding resolved at compile time.

RF links in this scenario

LINK	MISSION / ROLE	BAND	CHANNELS	BEHAVIOUR	RECORDING
L1	M1 · C2	2.4 GHz ISM	2412 / 2437 / 2462	scripted	urban_c2_2437.sigmf
L2	M1 · VIDEO	5.8 GHz	5745 / 5800	mission_triggered	video_dl_5800.sigmf
L3	M2 · TELEMETRY	915 MHz ISM	FHSS 50 ch	probabilistic_adaptive	telem_915_fhss.sigmf
L4	M2 · DATA	2.4 GHz ISM	2452	external_triggered	datalink_2452.sigmf
L5	M3 · VIDEO	5.8 GHz	5780	scripted	video_dl_5800.sigmf

Design notes

CHANNEL MODEL	Detection-only receivers need one signal per channel: the plan streams one drone per physical TX channel. Receivers that solve AOA or TDOA need parallel channels prepared together — one drone driven across a coherent group (one channel per array element or per TDOA node, with authored $\Delta\phi$ / τ offsets), combined on the RF network. A coherent group is allocated, leased, armed and muted as a single unit; a partial group is not a valid allocation.
PLANNED	Extension: combine multiple recordings at digital level and stream several drones per channel. The UI shape holds — a tile becomes a channel with N signal mappings instead of one, and detection-only allocations stop being 1:1 while coherent groups stay group-allocated.
ONE CLOCK	Development holds a single authoring scrub time T. The map evaluator, the Signal-panel playhead, and the Doppler sub-view all read the same T — nothing keeps its own cursor.
REPLAY ≠ DEV	Replay runs on run time from the run manifest, uses square red-accented run controls, and never exposes an authoring scrub — the two pages share evaluator code, not state.
EVENTS	Timeline events are the only writers of T besides the scrub itself — clicking an event seeks and pauses. Spatially-referenced events resolve their time from the trajectory, absolute ones are pinned.